

CEMI Career Database

Codebook v1.0

*Personnel, career-event, and demographic data for the Central Economic-Mathematical
Institute(CEMI / ЦЭМИ), USSR Academy of Sciences*

Archive: ARAN F.1959 / Op.1

Personnel-record coverage: 1961–1987 (27 calendar years)

Authors

To be completed by the project owner upon submission. This codebook is generated from the live build pipeline so that every variable definition matches what the latest `cemi_career.db` actually carries.

Purpose

The CEMI Career Database integrates the complete archival personnel record of the Central Economic-Mathematical Institute (CEMI) of the USSR Academy of Sciences with its surrounding archival metadata. It combines:

- individual-level career data for 1,954 persons reconstructed from 12,211 person-year observations across 1961–1987 (Career-side tables);
- aggregate demographic data from CEMI annual reports, including personnel totals, position breakdowns, academic-degree composition, nationality and age-bracket distributions, party membership, research field/subfield breakdowns, and research-trainee counts (Demographic-side tables);
- three layers of provenance — primary archival source descriptions from the ARAN F.1959 finding aid, the intermediate workbooks consumed by the build pipeline, and per-sheet OCR and English-translation pipeline records (Provenance tables);
- controlled vocabularies for institutions (with Nolting & Feshbach 1980/81 4-sector classification + a residual “Other” class), positions, degrees, fields of science, nationalities, age brackets, departments, schools, party organisations, and a 1,054-entry bilingual glossary.

The dataset supports quantitative study of Soviet academic-personnel dynamics — promotion timing, degree acquisition pathways, ethnic and party composition, transfer/dismissal flows, and the institutional ecology of late-Soviet economic-mathematical research.

Note on build tiers. The builder produces two tiers depending on which optional inputs are co-located with `data.xlsx` (README, §B.1). The full-fidelity build (56,931 total rows; 757 `sheets_index` rows; 1,054 glossary rows; 655 institution rows, 652 sector-classified) is the reference build documented throughout this codebook. The prebuilt `cemi_career.db` shipped at the repository root is the lean build (55,711 total rows; 181 `sheets_index` rows; 426 glossary rows; 639 institution rows, 636 sector-classified), produced from the three Excel inputs alone; all other 37 table counts are identical in both tiers. Tier-dependent row counts below are flagged where they occur.

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Data Units

The database is implemented as a single SQLite file, `cemi_career.db`, containing 41 tables. We organise them into five conceptual groups; each table is documented in §“Variable descriptions”.

Tables nest at three levels of analysis:

- Person — one row per unique individual (table `person`, 1,954 rows). Many tables hang off the `person_id` key.
- Person × Year — one row per (person, calendar year) observation (table `person_year_observation`, 12,211 rows). Captures position, department, specialty, degree, academic title, and grad/BA schools as observed in each annual record.
- Event — one row per discrete career event: appointment, dual appointment, transfer, dismissal, start of work, party membership, dissertation defense, academic-title award. Each event references `person_id` and most reference `calendar_year`.

Derived tables (§Methodological Notes 1, 2, 3) fold the raw event tables into analyst-friendly forms:

`cemi_position_span` (3,931 rows) collapses consecutive same-position appointments into ranges; `person_degree` (780 rows) groups observations into one row per (person, degree); `start_of_work.is_initial_join + span_id` link each start date to the position-span it opens.

Aggregate demographic tables (`demo_*`) operate at the (year, dimension) level — e.g. `demo_position` is keyed (year, `position_id`) and `demo_nationality` on (year, `nationality_id`).

Definitions of Key Terms

Person

A unique individual identified by `canonical_name` = "Surname Initial. Initial." OR (when the source uses the full Russian name) "Surname Given Patronymic". See §Methodological Note 1 for the patronymic-aware identity rule that prevents Filippova Nina Alekseevna and Filippova Natalya Arnoldovna from merging into a single Filippova N. A.

Position

A controlled-vocabulary entry from `position_rank` (47 distinct positions, with English label, Russian label where source-traceable, an institutional tier in {apprentice, junior, senior, head, leadership}, an ordinal `rank_order`, and a Boolean `is_acting` flag). Acting positions sit just below their substantive counterpart in `rank_order` so that within-year promotions “Acting $X \rightarrow X$ ” order correctly.

Position Span

A continuous stretch during which a person held the same (`position_id`, `is_acting`). Built by walking `cemi_appointment` in (year, `within_year_order`) order per person and folding consecutive rows that agree on (`position_id`, `is_acting`). See §Methodological Note 1.

Joined CEMI

The chronologically earliest `start_of_work` date per person, flagged with `is_initial_join` = 1. Subsequent `start_of_work` rows are promotion or rehire dates that correspond to second/third/... position-spans and are linked via `span_id` (Methodological Note 2).

Institutional Phase (CEMI-side axis)

The column `calendar_year.institute_phase` takes three values, but only two of them are substantive CEMI institutional phases. The two real phases — Predecessor Lab and CEMI — describe which Soviet institute existed at a given calendar year — the only fact this axis encodes; the person-side institutional classification is documented separately under Institutional Relation / Role. The third value, Pre-Institute, is a residual technical tag — not an institutional phase, and naming no institute — for calendar years (`year` < 1958) that pre-date even the predecessor laboratory and that appear in the database only because they are referenced by a person’s life event (party-membership year, BA-graduation date, etc.); no CEMI personnel record exists at such years. In the public UI the Career-Search Phase filter therefore exposes only the two real phases, while pre-1958 life-event dates remain searchable via the Active from / Active to year inputs.

- Predecessor Lab (1958–1962) — the Laboratory of Mathematical Methods Applied to Economic Research and Planning (Лаборатория экономико-математических методов АН СССР) out of which CEMI was reorganised in 1963.
- CEMI (1963–present) — the Central Economic-Mathematical Institute / Центральный экономико-математический институт АН СССР (ЦЭМИ).

Institutional Relation / Role (person-side axis)

Independently of the CEMI-side institutional phase above, every person carries up to four mutually orthogonal institutional-relation slots that classify the role they had towards a given institution. These four slots are encoded in four different tables/columns and are documented in §"Variable descriptions" below. Together they form the codebook's "role axis" — distinct from the institutional-phase axis — and the public UI exposes them as the Career-Search Role filter and the Institutions-tab CEMI-relationship filter.

- Primary Institution — `person_year_observation.primary_institution_id`. The person's primary employer at that calendar year, transcribed from `data.xlsx` column "Primary Institution". When this equals the CEMI row, the person is a full-time CEMI staff member that year; when it equals any other institution, that institution is the person's primary employer and CEMI is therefore only a part-time / dual post that year. Distribution in this release: 1,946 distinct persons with primary = CEMI (full-time) and 40 distinct persons with primary = external (CEMI = part-time) across 10 distinct external primary institutions (Predecessor Lab, Moscow State University, USSR State Planning Committee / Gosplan, CC CPSU, RSFSR Central Statistical Administration, Estonian Branch of CEMI, Tartu State University, etc.).
- Pre-CEMI Service — `pre_cemi_role.institution_id` (and `label` / `label_ru` / `kind`). External role held by the person BEFORE joining CEMI. By construction (Methodological Note 5) rows where the spreadsheet writes Previous Service Institution = "CEMI" / "ЦЭМИ" are excluded — they denote in-CEMI promotions captured by `cemi_position_span` instead. Coverage: 876 rows over 859 distinct persons.
- Dual / Part-time Appointment — `dual_position.institution_id` (with position, duration, etc.). Concurrent non-CEMI role held alongside the primary post. Coverage: 1,103 rows over 212 distinct persons.
- Transfer-out Destination — `transfer_event.institution_id` (with year and `institution_kind`). Destination of an outgoing transfer from CEMI. By the causal-order convention (Note 4) a transfer logically follows its same-year dismissal. Coverage: 557 rows over 548 distinct persons.

The two axes are independent: a single person-year observation can simultaneously sit in a particular phase (CEMI 1973) and carry any combination of role records (e.g. primary = CEMI, no pre-CEMI service, two dual posts that year, no transfer). An analytical query that conflates the two axes risks under- or over-counting. The public UI therefore documents them as two clearly separated filter axes and the codebook does the same.

Missing values

Two distinct missing-value conventions are used:

- NULL (the SQLite null) means the source-document did not record the value. This is the only convention used in any numeric/foreign-key column.
- "Don't Know" / "Unknown" / empty-string TEXT means the source-document explicitly marked the value as unknown. Used in event-detail fields such as `transfer_event.institution_kind`.

Data Entry Principles

The build pipeline observes six explicit principles, derived from the historical-database methodology literature — source-faithfulness and provenance preservation (Harvey & Press 1996, Ch. 5; Gil 2021, Ch. 2); persistent unique-identifier construction (Alter & Mandemakers 2014, IDS §3; Pasin & Bradley 2015, Factoid-based §2); event-based career observation with temporal causality (Mandemakers et al. 2020, Sowing Part II); and the distinction between "missing in source" and "explicitly unknown" (Foster & Godbole 2023, Database Systems: A Pragmatic Approach, Ch. 9 §9.2.2.2 — Codd's Rule 3 on the systematic treatment of null values):

1. Source-faithful transcription. Personnel records are entered exactly as they appear in the original ARAN F.1959 / Op.1 files. When a clear error is identified — typically by cross-referencing the same person across multiple years or against the institutional aggregate — the corrected information enters the database and the original or rejected reading is preserved in a Remarks / classification_evidence field.
2. Patronymic-distinct identity. Persons with the same surname and same Latin initials but different patronymics (Nina Alekseevna vs Natalya Arnoldovna) receive different person_id values. See Methodological Note 1.
3. Causal-order conservation. Within a single year, dismissal events render before transfer events, and within-year promotions sort by within_year_order so that an Acting position precedes its substantive counterpart. See Methodological Note 4.
4. In-institution promotion separation. When the source spreadsheet uses Previous Service Institution = "CEMI" / "ЦЭМИ" to encode an internal promotion, the row is not duplicated into pre_cemi_role; it is captured purely by the position spans. See Methodological Note 5.
5. Year-anchored defense routing. For persons holding both Ph.D. and National Doctor, each phd_defense row is routed to the person_degree row whose first_year matches the defense year. This prevents the National-Doctor-defense date from being attached to the Ph.D. row. See Methodological Note 6.
6. Provenance trail. Every demographic aggregate row carries a source_material reference back to the originating ARAN file (e.g. "Aran f. 1959, op. 1, d. 10(otchet)"), and every institution row carries a classification_evidence field linking the Nolting-sector assignment to the Russian-keyword or manual decision that drove it.

Data Files Overview

The complete release consists of a primary SQLite database, six input workbooks, and four pipeline folders of intermediate files. Reproducibility is guaranteed by re-running the two build scripts against the input files.

Primary outputs

File	Description	Approx. size
cemi_career.db	Main SQLite database — 41 tables	6 MB
cemi_career_ui.html	Single-file HTML viewer (Career Search, Provenance, etc.)	8 MB
CEMI_Codebook_v1.docx/.pdf	This document	~2 MB

Input workbooks

File	Role	Sheets / rows
data.xlsx	CEMI personnel records (one row per person-year observation)	12,211 × 27
cemi_translation_db.xlsx	Aggregate demographic + Source Files Index + Translation Glossary	11 sheets
research_field_subfield.xlsx	Subfield × year snapshots	22 sheets
institution glossary (Nolting 4-sector classified, all sheets).xlsx	Per-institution Nolting classifications + Type Codebook	5 sheets

Provenance pipeline folders

Folder	Content	Files
demographic db(new)/	Russian archival originals (1959 1 X отчет.xlsx)	23 files
OCR Results/	OCR-extracted Russian transcripts per archival file	25 files
English Tranlation Results/	English translations of OCR transcripts	25 files

Variable Descriptions — Career-side Tables

person

Row count: 1,954. One row per uniquely identified individual.

person_id	
Type	INTEGER (Primary key)
Value range	Synthetic 64-bit hash derived from canonical_name
Level	Person
Description	Unique identifier for the individual. Stable across rebuilds as long as canonical_name is unchanged.
Source	derive_personnel() — hash_id("person", canonical_name)

canonical_name	
Type	TEXT (NOT NULL, UNIQUE)
Value range	"Surname I. O." for initials-form sources; "Surname Given Patronymic" when the source spells the name fully
Level	Person
Description	Canonical display name. Constructed so that patronymic-distinct persons get distinct canonical_names (see Methodological Note 7).
Source	data.xlsx · "Name" column

surname	
Type	TEXT (NOT NULL)
Value range	Capitalised single token
Level	Person
Description	Family name in Latin transliteration.
Source	data.xlsx · "Name" column (parsed)

initials	
Type	TEXT

Value range	"I. O." form
Level	Person
Description	Abbreviated given-name and patronymic, always derived as "first letter + dot" regardless of how the source spells them out. Used by the UI for compact display in the Career Search results table.
Source	Derived from canonical_name

birth_year	
Type	INTEGER
Value range	1892–1961 in observed data
Level	Person
Description	Year of birth when recorded in any observation row for the person.
Source	data.xlsx · "Year of Birth" (first non-null observed)

ethnicity_id	
Type	INTEGER → ethnicity(ethnicity_id)
Value range	7 distinct ethnicities + NULL
Level	Person
Description	Ethnicity of the individual, captured at the person level (assumed constant within a person).
Source	data.xlsx · "Ethnicity"

person_name_variant

Row count: 1,954. Every distinct raw-name string observed for a person, together with the first calendar year that variant appeared.

variant_id	
Type	INTEGER (PK)
Value range	autoincrement
Level	Variant
Description	Synthetic key.
Source	derive_personnel()

person_id	
Type	INTEGER → person(person_id)
Level	Variant
Description	Person this variant refers to.

raw_name	
Type	TEXT (NOT NULL)
Value range	Verbatim source string
Level	Variant
Description	The exact "Name" cell content as it appears in data.xlsx. Persons normally have a single variant; persons whose source data switches between initials and fully-spelled forms have two.
Source	data.xlsx · "Name"

first_seen_year	
Type	INTEGER
Value range	1961–1987
Level	Variant
Description	Earliest calendar year in which this raw_name was observed.
Source	data.xlsx · "year"

person_year_observation

Row count: 12,211. One row per (person, year) observation in the source spreadsheet. Captures the institute snapshot for that person at that year — primary affiliation, department, specialty, degree, academic title, and grad/BA schools.

observation_id	
Type	INTEGER (PK)
Level	Observation
Description	Synthetic key.

person_id	
Type	INTEGER → person(person_id)
Level	Observation
Description	The person observed.

year	
Type	INTEGER → calendar_year(year)
Value range	1961–1987
Level	Observation
Description	Calendar year the source row applies to.
Source	data.xlsx · "year"

source_row_id	
Type	INTEGER → source_row(source_row_id)
Level	Observation
Description	Anchor back to the row index of the data.xlsx sheet that produced this observation.

primary_institution_id	
Type	INTEGER → institution(institution_id)
Level	Observation
Description	Primary employer at this year, transcribed from data.xlsx "Primary Institution". This column encodes the full-time vs. part-time distinction of the role axis (§"Institutional Relation / Role"): when it points to the CEMI institution row the person is full-time CEMI staff (1,946 distinct persons in this release); when it points to any other institution that institution is the person's primary employer and CEMI is correspondingly a part-time / dual post (40 distinct persons across 10 distinct external institutions, including the Predecessor Lab for pre-1963 rows, Moscow State University, USSR State Planning Committee, CC CPSU, etc.).
Source	data.xlsx · "Primary Institution"

department_id	
Type	INTEGER → department(department_id)
Level	Observation
Description	CEMI department, when recorded.
Source	data.xlsx · "department"

specialty_id	
Type	INTEGER → specialty(specialty_id)
Level	Observation
Description	Person's recorded specialty / major field for that year.
Source	data.xlsx · "specialty, major field"

degree_id	
Type	INTEGER → degree(degree_id)
Value range	Ph.D. or National Doctor or NULL
Level	Observation
Description	Degree the person holds at this year.
Source	data.xlsx · "Degree"

academic_title_id	
Type	INTEGER → academic_title(title_id)
Level	Observation
Description	Academic title (Professor, Docent, Academician, etc.) at this year.
Source	data.xlsx · "Academic Position"

grad_school_id	
Type	INTEGER → school(school_id)
Level	Observation
Description	Graduate school where the person earned their degree(s).

Source	data.xlsx · "Graduate School"
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ba_school_id	
Type	INTEGER → school(school_id)
Level	Observation
Description	School where the person earned their BA / Diplom.
Source	data.xlsx · "Graduate School of BA"

notes	
Type	TEXT
Level	Observation
Description	Free-text remarks copied from the source row.
Source	data.xlsx (residual cells)

cemi_appointment

Row count: 10,850. The raw appointment-events table: one row per (person, year) appointment to a CEMI position, including same-year promotions (Acting → substantive).

appointment_id	
Type	INTEGER (PK)
Level	Event
Description	Synthetic key.

person_id	
Type	INTEGER → person(person_id)
Level	Event
Description	Person appointed.

year	
Type	INTEGER → calendar_year(year)

Value range	1961–1987
Level	Event
Description	Calendar year of the appointment.
Source	data.xlsx · "year"

position_id	
Type	INTEGER → position_rank(position_id)
Value range	47 distinct positions
Level	Event
Description	Position the person is appointed to.
Source	data.xlsx · "Position" (cleaned)

rank_order	
Type	REAL
Value range	10–99
Level	Event
Description	Snapshot of the position's rank_order at insertion time (denormalised for fast UI sort).
Source	position_rank.rank_order

is_acting	
Type	INTEGER (0/1)
Level	Event
Description	1 if the appointment is an "Acting" variant (Acting Junior Researcher, Acting Senior Researcher, Acting Head of Sector, Acting Head of Laboratory).
Source	derived from position label

within_year_order	
Type	INTEGER
Value range	1–4

Level	Event
Description	Order within a calendar year — necessary because some persons have two or more appointments in the same year (e.g. Acting X → X promotion).
Source	assign_within_year_order()

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event
Description	Anchor back to the observation row that produced this event.

cemi_position_span — derived

Row count: 3,931. Each row is a continuous stretch during which a person held the same (position, is_acting).
 Folded from cemi_appointment by derive_career_artifacts(). Replaces year-by-year appointment rows in the UI Career Search profile view. See Methodological Note 1.

span_id	
Type	INTEGER (PK)
Level	Span
Description	Synthetic key.

person_id	
Type	INTEGER → person(person_id)
Level	Span

position_id	
Type	INTEGER → position_rank(position_id)
Level	Span
Description	Position held during the span.

is_acting	
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Type	INTEGER (0/1)
Level	Span
Description	Carried from cemi_appointment.

start_year	
Type	INTEGER
Value range	1961–1987
Level	Span
Description	First calendar year the (position, is_acting) was observed.

end_year	
Type	INTEGER
Value range	1961–1987
Level	Span
Description	Last calendar year the (position, is_acting) was observed.

n_years	
Type	INTEGER
Value range	1–23
Level	Span
Description	$\text{end_year} - \text{start_year} + 1$. Inclusive count.

span_order	
Type	INTEGER
Value range	1, 2, 3 ...
Level	Span
Description	1-based order within the person's career. $\text{span_order} = 1$ is, by convention, the “joined CEMI” span — the start_of_work date that opens this span carries $\text{is_initial_join} = 1$ (Note 2).

dual_position

Row count: 1,103 over 212 distinct persons. Records concurrent external (non-CEMI) appointments held alongside the primary CEMI post. This is the "Dual / part-time" slot of the role axis (§"Institutional Relation / Role").

dual_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

year	
Type	INTEGER → calendar_year(year)
Value range	1961–1987
Level	Event
Description	Calendar year the dual post was recorded.
Source	data.xlsx · "year"

institution_id	
Type	INTEGER → institution(institution_id)
Level	Event
Description	Outside institution where the dual post is held.
Source	data.xlsx · "Name of the institution where employed part-time"

position_label	
Type	TEXT
Value range	Russian: "доцент", "профессор", "Lecturer", "Editor", "Head of Departement", ...
Level	Event

Description	Position held externally. Free-text, not normalised against position_rank.
Source	data.xlsx · "position where employed part-time"

duration_text	
Type	TEXT
Value range	e.g. "1961", "1965-1966"
Level	Event
Description	Period during which the dual post was held, when reported.
Source	data.xlsx · "duration of the permitted part-time employment"

institution_kind	
Type	TEXT
Value range	{Academy, Branch, VUZ, Enterprise, Other}
Level	Event
Description	Nolting 5-sector classification of the institution. Mirrors institution.kind.
Source	institution glossary, Sheet2

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

pre_cemi_role

Row count: 876 over 859 distinct persons. External roles a person held before joining CEMI. This is the "Pre-CEMI service" slot of the role axis (§"Institutional Relation / Role"). By construction (Methodological Note 5) this table excludes rows where the spreadsheet uses Previous Service Institution = "CEMI" / "ЦЭМИ" — those denote in-CEMI promotions and are absorbed by cemi_position_span.

pre_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

institution_id	
Type	INTEGER → institution(institution_id)
Level	Event
Description	External institution.
Source	data.xlsx · "Previous Service Institution"

institution_kind	
Type	TEXT
Value range	{Academy, Branch, VUZ, Enterprise, Other}
Level	Event
Description	Nolting classification.
Source	institution glossary, Sheet1

title	
Type	TEXT
Value range	free-text
Level	Event
Description	Title held at the previous institution.
Source	data.xlsx · "Previous Institution Title"

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

transfer_event

Row count: 557 over 548 distinct persons. Outgoing transfers from CEMI to other institutions. This is the "Transfer-out destination" slot of the role axis (§"Institutional Relation / Role"). By the causal-order convention (Note 4), a transfer logically follows its accompanying dismissal within the same calendar year.

transfer_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

year	
Type	INTEGER → calendar_year(year)
Value range	1965–1987
Level	Event
Description	Year of the transfer.
Source	data.xlsx · "year"

institution_id	
Type	INTEGER → institution(institution_id)
Level	Event
Description	Destination institution.
Source	data.xlsx · "transferred institution"

institution_kind	
Type	TEXT
Value range	{Academy, Branch, VUZ, Enterprise, Other}
Level	Event

Description	Nolting classification.
Source	institution glossary, Sheet3

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

dismissal_event

Row count: 1,743. Dismissals from CEMI — usually carry a specific dismissal_date.

dismissal_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

year	
Type	INTEGER → calendar_year(year)
Value range	1963–1987
Level	Event
Source	data.xlsx · "year"

dismissal_date	
Type	DATE
Value range	"YYYY-MM-DD"
Level	Event
Description	Effective date of dismissal. When NULL only the year is known.

Source	data.xlsx · "date of dismissal"
---------------	---------------------------------

position_id	
Type	INTEGER → position_rank(position_id)
Level	Event
Description	Position the person was dismissed from.
Source	data.xlsx · "Position" at dismissal time

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

party_event

Row count: 285. Party membership events.

party_event_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

party_id	
Type	INTEGER → party_org(party_id)
Value range	CPSU or Komsomol
Level	Event
Description	Which party.
Source	data.xlsx · "Party Membership"

role	
Type	TEXT
Value range	{Member, Candidate}
Level	Event
Description	Role within the party — Member for full membership, Candidate for candidate (probationary) membership.
Source	data.xlsx · "Party Membership"

join_year	
Type	INTEGER
Value range	1940–1987
Level	Event
Description	Year of party membership acquisition.
Source	data.xlsx · "Year of Party Membership"

join_date	
Type	DATE
Value range	"YYYY-01-01"
Level	Event
Description	Synthetic ISO date constructed from join_year for use in time-axis charts.
Source	derived from join_year

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

phd_defense

Row count: 823. Dissertation defences. The source field is labelled "Dissertation defense date" and records both Ph.D. (Candidate of Sciences) and National Doctor (Doctor of Sciences) defences — see Methodological Note 6 for the routing rule that puts each defence on the correct degree.

defense_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

field_id	
Type	INTEGER → phd_field(field_id)
Value range	13 distinct fields
Level	Event
Description	Field of science of the dissertation.
Source	data.xlsx · "Field of Ph.D."

defense_date	
Type	DATE
Value range	"YYYY-MM-DD"
Level	Event
Description	Effective defence date.
Source	data.xlsx · "Dissertation defense date"

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

start_of_work

Row count: 3,460. A "Start Date of Work" row from the source spreadsheet — one per recorded promotion / role change / rehire date. After `derive_career_artifacts()` each row is linked to the position-span it opens (`span_id`) and

the earliest row per person is flagged with `is_initial_join = 1`.

start_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

start_date	
Type	DATE
Value range	"YYYY-MM-DD"
Level	Event
Description	The actual start date.
Source	data.xlsx · "Start Date of Work"

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

is_initial_join	
Type	INTEGER (0/1)
Level	Event
Description	1 = the chronologically earliest start row per person that maps to <code>span_order = 1</code> (the “joined CEMI” date). Persons whose original-hire row was not recorded carry 0 on every row.
Source	derive_career_artifacts()

span_id	
Type	INTEGER → cemi_position_span(span_id)

Level	Event
Description	Position-span this start date opens. Routing rules in Methodological Note 2.
Source	derive_career_artifacts()

academic_title_award

Row count: 11. Records the year a person was awarded a given academic title.

award_id	
Type	INTEGER (PK)
Level	Event

person_id	
Type	INTEGER → person(person_id)
Level	Event

title_id	
Type	INTEGER → academic_title(title_id)
Level	Event
Description	Title awarded.
Source	data.xlsx · "Academic Position" (when newly observed)

first_year	
Type	INTEGER
Level	Event
Description	First calendar year the title was observed.

observation_id	
Type	INTEGER → person_year_observation(observation_id)
Level	Event

person_degree — derived

Row count: 780. One row per (person, degree). Derived by `derive_career_artifacts()` from `person_year_observation` (first/last year a person appears with that degree) and back-filled from `phd_defense` (defence date + field) and `person_year_observation.grad_school_id` (school). Routing rules: Methodological Note 6.

person_id	
Type	INTEGER → person(person_id) — PK part 1
Level	Degree

degree_id	
Type	INTEGER → degree(degree_id) — PK part 2
Level	Degree
Description	Either Ph.D. or National Doctor.

first_year	
Type	INTEGER (NOT NULL)
Value range	1961–1987
Level	Degree
Description	First calendar year the person appears with this degree.
Source	<code>min(year) over person_year_observation</code>

last_year	
Type	INTEGER (NOT NULL)
Value range	1962–1987
Level	Degree
Description	Last such year.
Source	<code>max(year)</code>

n_years	
---------	--

Type	INTEGER (NOT NULL)
Value range	1–21
Level	Degree
Description	Number of distinct calendar years with this degree.
Source	COUNT(DISTINCT year)

school_id	
Type	INTEGER → school(school_id)
Value range	52 distinct schools
Level	Degree
Description	The graduate school where the degree was earned, as reported by the earliest observation with grad_school_id set.
Source	min(year) grad_school_id per (person, degree)

defense_date	
Type	DATE
Value range	"YYYY-MM-DD"
Level	Degree
Description	Effective dissertation defence date for this degree. Routed to the degree whose first_year matches the defence year; otherwise routed to Ph.D. by default. See Note 6.
Source	phd_defense (routed)

field_id	
Type	INTEGER → phd_field(field_id)
Value range	13 distinct fields
Level	Degree
Description	Field of science of the dissertation — populated for the same row defense_date is.
Source	phd_defense (routed)

Variable Descriptions — Demographic Aggregates

Aggregate demographic tables are loaded from the CEMI annual reports as they were re-organised in `cemi_translation_db.xlsx`. Most carry a `source_material` column documenting the originating ARAN file (e.g. "Aran f. 1959, op. 1, d. 10(otchet)").

demo_personnel_totals

Row count: 23. Top-line annual headcount.

year	
Type	INTEGER → <code>calendar_year(year)</code>
Value range	1961–1987
Level	Year
Description	Calendar year of the aggregate.

total_scientists	
Type	INTEGER
Value range	32–719
Level	Year
Description	Total number of scientific staff.
Source	<code>cemi_translation_db.xlsx</code> · Personnel Totals

total_all_staff	
Type	REAL
Value range	~37–1,107
Level	Year
Description	Total headcount including all staff categories.
Source	<code>idem</code>

national_doctor	
Type	REAL

Level	Year
Description	Number of National Doctor (Doctor of Sciences) holders.
Source	idem · "National Doctors"

phd

Type	REAL
Level	Year
Description	Number of Ph.D. (Candidate of Sciences) holders.
Source	idem · "Ph.D.s"

women

Type	REAL
Level	Year
Description	Number of women among the scientific staff.
Source	idem · "Women"

women_national_doctor

Type	REAL
Level	Year
Description	Women among National Doctor holders.
Source	idem

women_phd

Type	REAL
Level	Year
Description	Women among Ph.D. holders.
Source	idem

source_material

Type	TEXT
Value range	"Aran f. 1959, op. 1, d. XX(otchet)"
Level	Year
Description	Provenance — which ARAN file this row was compiled from.
Source	cemi_translation_db.xlsx · "Source Material"

demo_position

Row count: 98. Per (year, position) breakdown — aggregate counterpart of cemi_appointment.

year + position_id	
Type	composite PK
Level	Year × Position

total	
Type	INTEGER
Level	Year × Position
Description	Number of persons in that position that year.

national_doctor / phd	
Type	REAL
Level	Year × Position
Description	Degree breakdown within the (year, position) cell.

cpsu_members	
Type	REAL
Level	Year × Position
Description	CPSU members in that (year, position) cell.

komsomol	
----------	--

Type	REAL
Level	Year × Position
Description	Komsomol members in that cell.

source_material	
Type	TEXT
Level	Year × Position

demo_academic_degrees, demo_party

Row counts: 23 and 23. Annual academic-degree composition (with professors, docents, SNS title, academicians counts) and annual CPSU / Komsomol headcount + percentages. Both tables carry source_material. Full schema in cemi_career.db; the column meanings are identical to their cemi_translation_db.xlsx source sheets.

demo_nationality, demo_age

Row counts: 87 and 29. Per-(year, nationality) and per-(year, age bracket) breakdowns with total + degree composition. demo_age additionally carries 5 “academic title within age bracket” columns (academician_or_corr, professor, associate_professor, senior_researcher, junior_researcher) added in v3 of cemi_translation_db.xlsx. Year coverage of demo_age is sparse — 1966, 1972, 1977, 1982, 1987 — with 1987 using a finer 5-year bracket scheme (30–35, 36–40, ...).

demo_field, demo_subfield, demo_trainees

Row counts: 203, 812, 5. Research-field and research-subfield breakdowns (covering 1961–1987 with 22 distinct YYYY snapshots) and the annual trainee count. demo_field is derived from demo_subfield by aggregating subfields under each parent field.

Vocabulary Codebooks

institution

Row count: 655 (full-fidelity build; lean root build: 639). The institutional dimension. Each row carries a 5-sector kind classification per Nolting & Feshbach 1980/81 and, where available, the Russian-original label and the per-institution classification evidence.

institution_id	
Type	INTEGER (PK)
Value range	synthetic hash
Level	Institution

label	
Type	TEXT (NOT NULL, UNIQUE)
Value range	English label
Level	Institution
Description	Canonical English name.
Source	workbook (cleaned)

kind	
Type	TEXT
Value range	{Academy, Branch, VUZ, Enterprise, Other, NULL}
Level	Institution
Description	5-sector Nolting classification. See §"Institution kind taxonomy".
Source	institution glossary, Sheet1/2/3

label_ru	
Type	TEXT
Value range	Cyrillic, full form preferred
Level	Institution

Description	Russian-original name when available.
Source	institution glossary

classification_evidence	
Type	TEXT
Value range	free-text
Level	Institution
Description	Rationale for the kind assignment (Russian keyword match, manual decision, etc.).
Source	institution glossary

The institution.kind column takes one of five values: Academy / Branch / VUZ / Enterprise / Other. Sector definitions, operational criteria (Russian-keyword markers), per-sector primary sources, and the distribution in this release are presented in their own dedicated chapter, §"Institution Type Classification (Nolting 4-sector + Other)" — see the Table of Contents.

position_rank — the CEMI position ladder

Row count: 47. 47 distinct positions ordered from apprentice to leadership. rank_order is dimensionless — chosen so that "Acting X" sits just below the substantive "X" and so that the within-year promotion ordering Acting → Substantive is preserved.

Tier values:

- apprentice (6) — Trainee, Graduate Student, Junior Engineer, Engineer, Senior Engineer (also incl. Research Trainee).
- junior (9) — Lead Engineer, Leading Researcher, Leading Researchers, Acting Junior Researcher, Junior Researcher, Junior Researchers, Editor, Researcher, Senior Scientific And Technical Researcher.
- senior (17) — Art Editor, Chief Researchers, Economist, Other Researchers, Others, Researchers, Scientific-managerial Staff, Acting Senior Researcher, Senior Bibliographer, Senior Editor, Senior Researcher, Senior Researchers, Senior Researcher-consultant (+ -professor), Principal Researcher, Associate Professor, Acting Head Of Sector.
- head (11) — Head Of Sector, Acting Head Of Laboratory, Head Of Laboratory, Head Of Library, Head Of Subdivision, Deputy Head Of Department, Academic Secretary, Scientific Secretary, Chief, Head, Professor.

- leadership (4) — Deputy Director, Deputy Director Of The Laboratory, Director Of The Laboratory, Director.

Full ladder (rank_order ascending):

rank	English label	Russian (when known)	tier	acting?
10	Trainee		apprentice	
12	Research Trainee		apprentice	
14	Graduate Student		apprentice	
18	Junior Engineer		apprentice	
22	Engineer		apprentice	
26	Senior Engineer		apprentice	
30	Lead Engineer		junior	
30	Leading Researcher		junior	
30	Leading Researchers	ведущие научные сотрудники	junior	
33	Acting Junior Researcher		junior	✓
36	Junior Researcher		junior	
36	Junior Researchers	Младшие научные сотрудники	junior	
38	Editor		junior	
40	Researcher		junior	
43	Senior Scientific And Technical Researcher		junior	
50	Art Editor		senior	
50	Chief Researchers	главные научные сотрудники	senior	
50	Economist		senior	
50	Other Researchers	Прочие научные сотрудники	senior	
50	Others	Прочие	senior	
50	Researchers	научные сотрудники	senior	
50	Scientific-managerial Staff	Научно-руководящий персонал	senior	
52	Acting Senior Researcher		senior	✓
55	Senior Bibliographer		senior	

rank	English label	Russian (when known)	tier	acting?
55	Senior Editor		senior	
55	Senior Researcher		senior	
55	Senior Researchers	Старшие научные сотрудники	senior	
57	Senior Researcher-consultant		senior	
58	Senior Researcher-consultant, Professor		senior	
59	Principal Researcher		senior	
60	Associate Professor		senior	
62	Acting Head Of Sector		senior	✓
65	Head Of Sector		head	
67	Acting Head Of Laboratory		head	✓
70	Head Of Laboratory		head	
70	Head Of Library		head	
70	Head Of Subdivision		head	
73	Deputy Head Of Department		head	
75	Academic Secretary		head	
75	Scientific Secretary		head	
77	Chief		head	
77	Head		head	
80	Professor		head	
92	Deputy Director		leadership	
95	Deputy Director Of The Laboratory		leadership	
95	Director Of The Laboratory		leadership	
99	Director		leadership	

degree, party_org and party_event roles

Both tables are tiny controlled vocabularies that anchor the personnel-side analyses.

degree.label	Russian equivalent	Soviet usage notes
Ph.D.	Кандидат наук	Candidate of Sciences (kandidat nauk) — the first dissertation degree. Defended after ~3 years aspirantura.
National Doctor	Доктор наук	Doctor of Sciences (doktor nauk) — the higher dissertation degree, typically earned a decade or more later.
party_event observations		Count
CPSU — Member		199
CPSU — Candidate		84
Komsomol — Member		2

ethnicity, nationality, age_bracket, phd_field, academic_title

Smaller controlled vocabularies used by individual or aggregate tables.

ethnicity (person-level)

label_en
Armenian
Estonian
Jewish
Kabardian
Korean
Russian
Ukrainian

nationality (year-level demographic)

label_ru	label_en
Армане	Armenians
Белорусы	Belarusians
Эстонцы	Estonians
Грузины	Georgians
Евреи	Jews
Корейцы	Koreans
Латыши	Latvians

label_ru	label_en
Осетины	Ossetians
Русские	Russians
Татары	Tatars
Украинцы	Ukrainians

age_bracket

label_en	label_ru	lower	upper
up to 29 years (inclusive)	до 29 лет(включительно)		29
61 years and older	61 год и старше		
30–40 years	30-40 лет	30	40
30–35 years	30-35 лет	30	35
36–40 years	36-40 лет	36	40
41–50 years	41-50 лет	41	50
41–45 years	41-45 лет	41	45
46–50 years	46-50 лет	46	50
51–60 years	51-60 лет	51	60

phd_field

label
Agricultural Sciences
Biological Sciences
Chemistry
Economics
Engineering
Geography
Geology
History
Mathematical Economics
Medical Sciences
Philosophy

label
Physics and Mathematics
Technical Sciences

academic_title (by rank_order descending)

label_en	rank_order
Academician	100
Corresponding Member of the USSR Academy of Sciences	80

glossary — bilingual term reference

Row count: 1,054 (full-fidelity build; lean root build: 426). A 4-column reference table (russian, english, category, glossary_id) supplying Russian↔English equivalents for every controlled term in the database.

Categories actually present:

category	count
institution	628
field_of_science	133
other	75
form_label	62
nationality	54
position	31
gender_age	26
degree	14
personnel_count form_label	12
institution_type	5
gender_age form_label	5
field_of_science form_label	3
party	2
degree field_of_science form_label	2
personnel_count field_of_science form_label	1
gender_age institution	1

The institution category is the largest, and together with institution_type they supply the 623 distinct institutions classified in the Nolting glossary workbook and the 5 sector definitions of the Type Codebook.

Institution Type Classification (Nolting 4-sector + Other)

Every institution in the CEMI Career Database carries one of five sector labels: Academy, Branch, VUZ, Enterprise, or Other. The four substantive sectors (Academy, Branch, VUZ, Enterprise) are the canonical Soviet R&D sector taxonomy introduced by Nolting & Feshbach (1980, 1981) on the basis of the TsSU (Central Statistical Administration) Science & Technology statistics universe and refined for analytic use by subsequent secondary literature (Graham 1975; Cocks 1980). The "Other" sector is a residual category capturing institutions outside the TsSU S&T universe (libraries, publishing houses, hospitals, schools, military units, etc.).

Coverage in this release: 655 distinct institution rows in the institution table (full-fidelity build). The kind column was back-filled for 652 of them by the Type Codebook + Sheets 1/2/3 of the Nolting institution glossary workbook; the remaining 3 carry NULL (no classification was attempted because the institution did not appear in any of the three sheets). In the lean root build, which skips the glossary enrichment, the corresponding figures are 639 institution rows, 636 of them classified by the name-based cascade alone.

Each sector is defined below at three levels: conceptual definition (what an institution must be); operational criterion (the Russian/English keyword markers used during coding); and the primary-source citations supporting the boundary.

Academy — академический сектор

Conceptual definition. Institutions of the USSR Academy of Sciences (АН СССР) and its 14 territorial branches, the 14 republic academies of sciences, and the four specialised academies — Academy of Medical Sciences (АМН), V. I. Lenin All-Union Academy of Agricultural Sciences (ВАСХНИЛ), Academy of Pedagogical Sciences (АПН), and (from 1976) Academy of Arts. Defining feature: the institution is administratively subordinated to one of these academies.

Operational criterion. The institution name contains one of the markers «АН», «Академия наук», «АМН», «ВАСХНИЛ», «АПН», or the English equivalent ("Academy of Sciences", "Academy of Medical Sciences", "Academy of Pedagogical Sciences", "VASKhNIL"), and the institution is not a higher-education establishment that merely uses the word "Academy" in its name (e.g. military VUZ-equivalent academies → VUZ, see below).

Primary sources. Nolting & Feshbach (1980), *Science* 207(4430), p. 494, Table 1 "academic sector"; Nolting & Feshbach (1981), Bureau of the Census P-95 No. 76, §II.A, Table 2.

Branch — отраслевая наука

Conceptual definition. Sectoral / "ministry-line" research institutes that conduct R&D in support of a particular production ministry, state committee, or main administration (главк), without being part of either the Academy system or the higher-education system. Encompasses: scientific research institutes (НИИ / ЦНИИ / ВНИИ / ОНИИ), design bureaus (КБ / ОКБ / ЦКБ / СКБ), state design-survey institutes (ГСПИ / ГПИ), scientific-

production associations (НПО), and trusts subordinated to a ministry or state committee.

Operational criterion. The institution name contains a ministerial or state-committee tag («Министерство», "USSR State Planning Committee", "Госплан", "Госкомтруд", "ГКНТ", «главк», «трест»), OR a sector-research marker («НИИ», «ЦНИИ», «ВНИИ», «ОНИЛ», «КБ», «ОКБ», «ЦКБ», «СКБ», «ГСПИ», «ГПИ», «НПО»), AND no higher-education or academic-sector marker (no «АН», «университет», «институт народного хозяйства», etc.).

Primary sources. Nolting & Feshbach (1980), *Science* 207(4430), pp. 494–495, Table 1; Nolting & Feshbach (1981), Bureau of the Census P-95 No. 76, §II.B, Tables 3–5; Decree of the USSR Council of Ministers No. 760, 24 September 1968, "On the establishment of production and scientific-production associations" (formalising НПО status as part of the Branch sector); Cocks (1980), NSF Science Policy in the USSR, Ch. VIII.

VUZ — вузовский сектор

Conceptual definition. Higher-education institutions performing R&D as part of their teaching mission. Encompasses: universities (университеты), polytechnic institutes (политехнические), specialist higher-education institutes by professional domain (педагогический, медицинский, финансовый, инженерно-экономический, etc.), the Academy of Foreign Trade, the Academy of National Economy (АНХ), the Academy of Social Sciences under the CC CPSU (АОН при ЦК КПСС), agricultural academies, military VUZ-equivalent academies, IPK (institutes for advanced training / Институты повышения квалификации), and tekhnikums (техникумы).

Operational criterion. The institution name contains an educational-establishment marker: «университет», «политехнический», «педагогический», «медицинский», «финансовый», «институт народного хозяйства», «инженерно-экономический», «ИПК», «техникум», «высшее училище», or "School of" / "Higher School of" / "University" in English; OR is one of the recognised Moscow-area named-after specialist higher-education institutes (МИИТ, МИСи, МИКХМ, МИЕТ, МГУ, МГПИ, МЕИС, etc.). Military "academies" that are functionally higher-education institutes (Frunze, Lenin Military-Political, Higher Combined Arms Command School, etc.) are coded as VUZ on this criterion.

Primary sources. Nolting & Feshbach (1980), *Science* 207(4430), p. 495, Table 1; Graham (1975), "The Formation of Soviet Research Institutes: a combination of revolutionary innovation and international borrowing", *Social Studies of Science* 5(3), § "The educational pyramid"; Nolting & Feshbach (1981), Bureau of the Census P-95 No. 76, §II.C, Table 6; Decree of the USSR Council of Ministers of 13 June 1961, "On measures for the further development of scientific-pedagogical cadres in higher educational institutions of the USSR" (formalising VUZ research-funding stream).

Enterprise — заводская наука

Conceptual definition. R&D, design, and technological subdivisions embedded inside a production enterprise —

i.e. their parent is a factory / combine / production association whose primary output is goods or services, and the R&D unit exists to serve that production line. Encompasses: research and design subdivisions of заводы, фабрики, комбинаты, производственные объединения, тресты, автокомбинаты; central topographic enterprises; shipyards; railway operating divisions (отд. ж.д.); and similar.

Operational criterion. The institution name contains a production-enterprise marker: «завод», «фабрика», «комбинат», «производственное объединение», «трест», «автокомбинат», «отд. ж.д.», «верфь», or the English equivalent ("Plant", "Factory", "Combine", "Production Association", "Trust", "Shipyard").

Distinguished from Branch by the parent — a Branch institute is subordinated to a ministry; an Enterprise R&D unit is subordinated to a production entity.

Primary sources. Nolting & Feshbach (1980), *Science* 207(4430), pp. 495–496, Table 1; Nolting & Feshbach (1981), Bureau of the Census P-95 No. 76, §II.D; Cocks (1980), NSF Science Policy in the USSR, Ch. VIII §3.

Other — residual / outside the TsSU S&T universe

Conceptual definition. Bodies that fall outside the four-sector TsSU S&T statistics universe altogether: libraries, publishing houses, journals, museums, hospitals and sanatoria, sports societies, banks that do not finance R&D, foreign organisations, post offices, primary and secondary schools, military units of undisclosed function (войсковая часть, в/ч), and post-box (п/я) entities of undetermined sector.

Operational criterion. Negative: no Russian/English keyword matches any of the Academy / Branch / VUZ / Enterprise markers above, OR the institution carries an explicit non-R&D tag (журнал, музей, библиотека, больница, школа №...).

Primary source. Nolting & Feshbach (1981), Bureau of the Census P-95 No. 76, §I "Coverage and exclusions".

Distribution in this release

Per the institution table (655 rows total):

Sector	Russian sector name	Institutions	Share
Branch	отраслевая наука	342	52.2%
VUZ	вузовский сектор	111	16.9%
Academy	академический сектор	85	13.0%
Other	— (residual)	76	11.6%
Enterprise	заводская наука	38	5.8%
(NULL)	—	3	0.5%

Per-institution evidence

For every institution that received a sector assignment, the rationale is preserved in the `institution.classification_evidence` column. Three patterns are observed in the source workbook:

- Russian-keyword match. Format: `RU: '<keyword>'`. E.g. "RU: 'НИИ'" → Branch; "RU: 'АН'" → Academy; "RU: 'Министерств'" → Branch. Applied automatically by the Nolting-glossary coding stage when the Russian short-form contains the keyword.
- English-keyword match. Format: `EN: '\b<keyword>\b'`. E.g. "EN: '\bAcademy of National Economy\b'" → VUZ. Applied when the English name unambiguously identifies the institution sector.
- Manual override. Format: `Manual: <free-text rationale>`. Applied to ambiguous cases that require human judgment, e.g. "Manual: AON pri TsK KPSS — Academy of Social Sciences (VUZ-equivalent)" (an "Academy" by name but a VUZ by function).

Analysts can therefore audit every classification with a single SQL query:

```
SELECT label, label_ru, kind, classification_evidence FROM institution WHERE kind = 'VUZ' AND classification_evidence LIKE 'Manual:%' ORDER BY label;
```

Edge cases and consistency notes

- VUZ-equivalent academies. Institutions whose name contains «Академия» but whose function is higher-education-equivalent (Academy of Foreign Trade, Academy of National Economy, AON pri TsK KPSS, Academy of the USSR Ministry of Internal Affairs, etc.) are coded VUZ, not Academy. The `classification_evidence` string carries a manual override identifying the case.
- CEMI internal branches. CEMI's own Estonian Branch and Leningrad Branch are coded Academy (because they are subordinated to the USSR Academy of Sciences via CEMI), not Branch — despite the English word "Branch" in their names.
- Pre-1968 Production Associations. Institutions classified as NPO under the 24 September 1968 decree are coded Branch; the corresponding production-side units (parent factories) are coded Enterprise. A few transitional cases (1965–1968) are coded by primary mission as recorded in the source.
- Unclassified rows. Institutions appearing only in the personnel data (`data.xlsx`) but not in any sheet of the Nolting glossary workbook carry `kind = NULL` and `classification_evidence = NULL`. In this release the residual NULL count is 3 out of 655.

References specific to this classification

Primary taxonomic sources (4-sector framework)

- Nolting, L. E. & Feshbach, M. R&D Employment in the U.S.S.R. *Science* **207**, 493–503 (1980). — concise statement of the Academy / Branch / VUZ / Enterprise taxonomy.

- Nolting, L. E. & Feshbach, M. *Statistics on Research and Development Employment in the U.S.S.R.* Series P-95, No. 76 (U.S. Bureau of the Census, 1981). — book-length elaboration; §I–IV define per-sector universe boundaries.
- Nolting, L. E. *Financing of Research, Development, and Innovation in the U.S.S.R.* Foreign Economic Report No. 22 (U.S. Bureau of the Census, Washington, D.C., 1985). — financing-side definition of the Branch / Enterprise distinction.

Soviet-side primary sources (TsSU methodology + legal framework)

- USSR Central Statistical Administration (ЦСУ СССР). *Народное хозяйство СССР за 60 лет / 1977: методология* (Statistika, 1977). — official TsSU statistical-methodology yearbook defining the Soviet S&T statistical universe that the Nolting taxonomy operationalises.
- USSR Central Statistical Administration (ЦСУ СССР). *Народное образование, наука и культура в СССР: статистический сборник* (Statistika, 1977). — VUZ-sector and scientific-research-establishment statistics for the period 1965–1976.
- USSR Central Statistical Administration (ЦСУ СССР). *Народное хозяйство СССР: технический прогноз* (Statistika, 1977/1978). — Soviet-side documentation of R&D employment by sector, used to cross-check the Nolting coverage.
- USSR Council of Ministers. Decree No. 760 of 24 September 1968 «О создании производственных и научно-производственных объединений» (On the establishment of production and scientific-production associations). — formalises the НПО status that pins the Branch / Enterprise boundary.
- USSR Council of Ministers. Decree of 13 June 1961 «О мерах по дальнейшему развитию научно-педагогических кадров в высших учебных заведениях СССР» (On measures for the further development of scientific-pedagogical cadres in higher educational institutions of the USSR). — formalises the VUZ-sector R&D funding stream.
- CPSU Central Committee Plenum. Resolution on the development of the USSR economy (with the attendant restructuring of научно-технические institutions), November 1962. — political-administrative context for the early-1960s wave of branch-institute formation (including CEMI's 1963 reorganisation).

Western secondary sources (sector-by-sector commentary)

- Graham, L. R. The formation of Soviet research institutes: a combination of revolutionary innovation and international borrowing. *Social Studies of Science* 5, 303–329 (1975). — historical basis for the VUZ-sector treatment of higher-education research.
- Cocks, P. M. *Science Policy in the Soviet Union*. Science Policy: USA/USSR, Vol. II, Report NSF/PRA-NS-80-09 (National Science Foundation, 1980). — Ch. VIII analyses ministry-line ("Branch") R&D organisation.

Defence-industry / specialist-sector references (boundary cases)

- Kassel, S. & Campbell, C. *The Soviet Academy of Sciences and Technological Development*. Report R-2533 (RAND Corporation, Santa Monica, CA, 1980). — reference for Academy / Branch boundary cases in defence-related R&D.
- Central Intelligence Agency. *Organization of Soviet Science*. Report CIA/RR ER-59-22 (CIA-RDP79R00904A000500010075-6) (1959). — earliest English-language sector-by-sector inventory of Soviet R&D institutions.
- Gerovitch, S. *Voices of the Soviet Space Program* — Glossary of Soviet space-program institutions (RU/EN) (Palgrave Macmillan, 2014). — supplementary glossary for branch / enterprise classification of space-sector institutes (КБ / ОКБ / ЦКБМ).
- Dexter, K. & Rodionov, I. *The Soviet Defence Industry Enterprise Database* (Warwick CSGR); consolidated workbook + per-entity preface PDFs (2007–2017). — primary reference for НИИ / КБ / ОКБ / ЦКБ / СКБ / ГСПИИ classifications in the defence-industry corpus.

Variable Descriptions — Provenance Tables

sheets_index

Row count: 757 (full-fidelity build; lean root build: 181). Per-sheet provenance. Every Excel sheet that participates in the build — archival, intermediate, OCR, English-translation, or personnel — gets one row.

Filters: role and target_table.

year	
Type	INTEGER
Value range	1961–1987
Level	Sheet
Description	Year the sheet covers (when extractable from filename / sheet name).

institution	
Type	TEXT
Value range	"Central Economic-Mathematical Institute (CEMI), USSR Academy of Sciences" / "Laboratory of Mathematical Methods..."
Level	Sheet
Description	Recorded for 712 of the 757 sheet rows (full-fidelity build).

source_file	
Type	TEXT
Value range	*.xlsx filename
Level	Sheet
Description	The workbook the sheet belongs to.

sheet_name	
Type	TEXT
Level	Sheet
Description	Sheet tab name. For pipeline files, the suffix "[→ 1959_1_N_отчет.xlsx]" links to the archival original by file-number.

first_cell_ru / first_cell_en	
Type	TEXT
Level	Sheet
Description	Cell A1 content (Russian and/or English) when available.

rows / cols	
Type	INTEGER
Level	Sheet
Description	Sheet dimensions.

role	
Type	TEXT
Value range	{English translation, OCR intermediate, primary archival source, intermediate workbook, personnel workbook}
Level	Sheet
Description	Pipeline role of the sheet. Distribution: English translation=294, OCR intermediate=282, primary archival source=147, intermediate workbook=33, personnel workbook=1.

target_table	
Type	TEXT
Level	Sheet
Description	Hand-curated mapping to the DB table the sheet feeds (e.g. "Positions" → demo_position, "1965.01" → demo_subfield).

source_row, calendar_year

source_row (12,211 rows) anchors each personnel observation back to the (sheet, row_index) location in data.xlsx. calendar_year (27 rows) maintains a single row per year present anywhere in the dataset, carrying institute_phase and institute_label for that year (Pre-Institute / Predecessor Lab / CEMI).

Methodological Notes

Note 1 — Position-span consolidation

For every (person, year) appointment, the build pipeline walks `cemi_appointment` in (year, within_year_order) order per person and folds consecutive rows that agree on (position_id, is_acting) into a single span (start_year, end_year, n_years).

Concrete example. Altaev V. Ya. has 17 appointment rows folded into 5 spans. His 1965 record carries two appointments: row 1 Senior Engineer (within_year_order = 1), row 2 Acting Junior Researcher (within_year_order = 2). These become two single-year spans (span_order = 1 and 2). His 1966–1969 four Lead Engineer rows collapse into one span (start_year = 1966, end_year = 1969, n_years = 4, span_order = 3).

Note 2 — start_of_work routing and is_initial_join

Each `start_of_work` row is the start date of one position-span. `derive_career_artifacts()` walks every start row and routes it via a tiered match against `cemi_position_span`:

tier-1 EXACT — the `start_date` year equals a span's `start_year`. Multiple same-year start rows are paired with multiple same-year spans by date order ↔ `span_order`: the i-th start in the year goes to the i-th span starting that year.

tier-2 PRE-CEMI — the `start_date` year falls before the earliest span's `start_year` (the “joined the predecessor laboratory before formal CEMI records begin” case). Routes to `span_order = 1`.

tier-3 MID-SPAN — the `start_date` year falls strictly inside a span ($\text{start_year} < y \leq \text{end_year}$). Routes to the containing span; carries `is_initial_join = 0`.

`is_initial_join = 1` is set on the chronologically-earliest `start_of_work` row per person that routes to `span_order = 1`. Among 1,954 persons, 1,718 have at least one recorded `start_date`, and 1,679 carry exactly one row flagged `is_initial_join = 1`. The remaining 236 persons have no recorded `start_date` and carry 0 on every row.

Note 3 — person_degree back-filling

`person_degree` starts as the grouping (MIN(year), MAX(year), COUNT(DISTINCT year)) over `person_year_observation` per (person_id, degree_id) — yielding 780 rows.

Then `derive_career_artifacts()` back-fills three optional columns: `school_id` is the earliest non-NULL `grad_school_id` observed for that (person, degree); `defense_date` and `field_id` come from `phd_defense` routed by the rule of Note 6.

Coverage: 290 rows have `school_id` (~37%), 387 have `defense_date` (322 Ph.D. + 65 National Doctor — see Note 6

for why the Ph.D. share is lower than at first glance), 712 have field_id.

Note 4 — Causal ordering: Dismissal before Transfer

When a person has both a dismissal and an outgoing transfer in the same calendar year (550 person-year pairs across 541 persons), the UI timeline renders dismissal first, transfer second. Rationale: the person is dismissed from CEMI (leaves), then is transferred to a destination institution (arrives).

Example: Svetlov K. S. 1980 — dismissed from Acting Senior Researcher on 1980-11-17, transferred to the State Research Institute of Power Engineering named after G. M. Krzhizhanovskii (same year).

Implementation: openPerson() in the UI assigns ord = 9 to dismissal events and ord = 9.5 to transfer events so that any same-year tie sorts in the right direction. The DB does not encode an intra-year order column on these tables; the rule lives entirely in the rendering pipeline and is documented in an SQL comment on the transfer_event CREATE TABLE.

Note 5 — Pre-CEMI in-institution filter

The personnel spreadsheet uses Previous Service Institution = "CEMI" to encode an in-CEMI promotion — e.g. row carries Position = "Junior Researcher", Previous Service Institution = "CEMI", Previous Institution Title = "Trainee".

These rows are not duplicated into pre_cemi_role because the position transition they represent is already captured by cemi_position_span. load_personnel() skips the insertion when pre_inst.upper() ∈ {"CEMI", "ЦЭМИ"}.

After this filter pre_cemi_role contains 876 rows of genuine external prior service. Top pre-CEMI institutions: Moscow State University (70), Institute of Economics of the USSR Academy of Sciences (23), Economic Research Institute of the USSR State Planning Committee (20), Moscow Institute of Physics and Technology (18), Soviet Army and Council for Study of Productive Forces under Gosplan (15 each).

Note 6 — Defense-date routing across Ph.D. and National Doctor

The personnel spreadsheet has a single "Dissertation defense date" column that records both Ph.D. and National Doctor defences. A naïve attribution of every phd_defense row to the Ph.D. person_degree row would mis-route 46 persons (those holding both degrees) by attaching their National Doctor defence date to their Ph.D. row.

The build pipeline routes each phd_defense row by matching its defence YEAR to person_degree.first_year for that person. When the defence year equals a particular degree's first_year, that is a strong signal that the defence was for THAT degree.

Example: Aivazyan S. A. has Ph.D. first_year = 1969, National Doctor first_year = 1976, and a single phd_defense row dated 1976-12-10. The router attaches 1976-12-10 to the National Doctor row, leaving Ph.D.defense_date = NULL. Both rows preserve the field "Physics and Mathematics".

Verified outcome: 0 cases remain where a Ph.D. row carries a defence date equal to the National Doctor first_year.

Note 7 — Patronymic-aware person identity

Persons whose surname and Latin initials coincide but whose full patronymic differs receive distinct person_id values. Without this rule, four pairs collapse into single rows in earlier builds: Filippova Nina Alekseevna + Filippova Natalya Arnoldovna (both initial-form "N. A."); Goldenberg A. I. + Goldenberg Aleksei I.; Leontieva L. L. + Leontieva Lyu. L.; Rodionov M. A. + Rodionov M. Aleksandrovich.

Implementation in split_name(): a token of the raw "Name" cell is considered "spelled" when its alphabetic core is ≥ 3 characters. If any rest-token (after the surname) is spelled, canonical_name is "Surname Token1 Token2 ..." verbatim; otherwise canonical_name takes the initials form "Surname I. O.". The initials column is still derived consistently in initials form for compact UI display.

Coding Examples

Worked walk-throughs of two persons whose data exercises every methodological rule above.

Example 1 — Aivazyan S. A. (Note 6 — defense routing)

person_id 265632827644960. Holds both Ph.D. (1969–) and National Doctor (1976–). Single phd_defense row, defence date 1976-12-10, field “Physics and Mathematics”.

person_degree rows after derive_career_artifacts():

first_year	last_year	degree	defense_date	field	school
1969	1975	Ph.D.	— (NULL)	Physics and Mathematics	—
1976	1987	National Doctor	1976-12-10	Physics and Mathematics	—

The 1976-12-10 defence routes to the National Doctor row because the defence year matches that degree’s first_year. The Ph.D. row retains its field but carries no defence date (the source did not record the Ph.D. defence date).

Example 2 — Altaev V. Ya. (Notes 1 & 2 — span pairing)

person_id 127607000466114. Five career spans covering 1965–1979, with two same-year spans in 1965.

cemi_position_span:

span_order	start_year	end_year	position	is_acting
1	1965	1965	Senior Engineer	
2	1965	1965	Acting Junior Researcher	✓
3	1966	1969	Lead Engineer	
4	1969	1969	Acting Senior Researcher	✓
5	1970	1979	Senior Researcher	

start_of_work (with span_id linkage):

start_date	is_initial_join	span_id (→ span_order)
1965-02-02	1	1756
1965-04-01	0	1757
1966-07-16	0	1758
1969-12-16	0	1759

start_date	is_initial_join	span_id (→ span_order)
1970-07-02	0	1760

1965-02-02 routes to span_order = 1 (Senior Engineer); 1965-04-01 routes to span_order = 2 (Acting Junior Researcher) — both via tier-1 EXACT match. The chronologically earlier 1965-02-02 row is flagged is_initial_join = 1. Subsequent rows (1966-07-16, 1969-12-16, 1970-07-02) route to spans 3, 4, 5 respectively.

Source Materials (Pipeline File Inventory)

Each archival original (1959 1 X отчет.xlsx) sits at the top of a four-stage transformation pipeline. We document the full inventory in order to satisfy the source-to-database traceability requirement that the historical-DB methodology literature treats as essential to reproducibility (Harvey & Press 1996, Ch. 4 "Source criticism and data capture"; Alter & Mandemakers 2014, IDS §2 on raw-source preservation; Gil 2021, Ch. 1 §1.3 on documenting the chain from archive to relational schema).

Stage 0 — Primary archival sources (ARAN F.1959 / Op.1)

Every record in cemi_career.db derives ultimately from a single archival series — the Archive of the Russian Academy of Sciences (ARAN), Fond 1959 (Central Economic-Mathematical Institute), Opis 1 (management documentation). The table below is the complete inventory of the 26 archival files (dela) consulted — 1,225 sheets in total, covering 1961–1988 — from which the Stage-1 Russian workbooks documented below were transcribed. The Track A and Track B columns give each file's access status at the time of consultation (Open = consulted; Sealed = not released under ARAN's access policy); these are the origin of the coverage gaps documented in §“Data Coverage and Gaps”. The full Russian titles and the per-file statistical-form lists are provided in the deposited inventory ARAN_F1959_Op1_CEMI_inventory.xlsx.

Delo №	Year(s)	Document (archival file title)	Sheets	Track A	Track B
10	1961	Central Economic-Mathematical Institute. Annual report and statistical data on personnel records for 1961 (forms 5-nk, 9). Laboratory for the Application of Mathematical Methods in Economic Research.	22	Open	Open
19	1962	(Central Economic-Mathematical Institute). Laboratory for the Application of Mathematical Methods in Economic Research and Planning. Annual reports and statistical data on personnel records for 1962 (forms 5-nk, 9, 9-sr, 10-nr).	30	Open	Open
30	1963	Annual reports and statistical data on personnel records of the Central Economic-Mathematical Institute for 1963 (forms 5-nk, 9, 9-sr, 10-npr, 2).	33	Open	Open
48	1964	Annual report and statistical data on personnel records of the Central Economic-Mathematical Institute for 1964 (forms 5-nk, 9, 9-sr, 10-npr, 10-t).	26	Open	Open
76	1965	Annual reports and statistical data on personnel records of the Central Economic-Mathematical Institute for 1965 (forms No. 5-nk, 2/5-nt/urgent, 9, 9-sr, 10-t, 10-npr).	38	Open	Open
111	1966	Annual report and statistical data on personnel records of the Central Economic-Mathematical Institute for 1966 (forms 5-nk, 10-t, 9, 9-sr).	62	Open	Open
152	1967	Annual report and statistical data of the Central Economic-Mathematical Institute on personnel records for 1967 (forms 5-nk, 10-t, 13-t3).	45	Open	Open
186	1968	Annual report and statistical data on personnel records of the Central Economic-Mathematical Institute for 1968 (forms 5-nk, 9, 9-sr).	59	Open	Open
222	1969	Annual report and statistical data on personnel records of the Central Economic-Mathematical Institute for 1969 (forms 5-nk, 10-t).	42	Open	Open
261	1970	Annual report and statistical data of the Central Economic-Mathematical Institute on personnel records for 1970 (forms 5-nk, 10-t, 9-sr, 9).	58	Open	Open
306	1971	Annual reports and statistical data of the Central Economic-Mathematical Institute on personnel records for 1971 (forms 5-nk, 1-migration, 6-t).	45	Open	Open
352	1972	Annual reports and statistical data on personnel records of the Central Economic-Mathematical Institute for 1972 (forms 5-nk, 6-t).	39	Open	Open
405	1973	Annual reports and statistical data on personnel records for 1973 (forms 5-nk, 9, 9-sr,	51	Open	Open

		6-t, 2).			
453	1974	Annual reports and statistical data on personnel records of the Central Economic-Mathematical Institute for 1974 (forms 5-nk, 6-t).	37	Open	Open
512	1975	Annual reports and statistical data on personnel records of the Central Economic-Mathematical Institute for 1975 (forms 5-nk, 9-sr, 9).	52	Open	Open
569	1976	Annual reports and statistical data on personnel records of the Central Economic-Mathematical Institute for 1976 (form 5-nk).	38	Open	Open
619	1977	Annual reports and statistical data on personnel records of the Central Economic-Mathematical Institute for 1977 (forms 5-nk, 6-t, 9, 9-sr, 2).	50	Open	Open
685	1978	Annual report and statistical data on personnel records for 1978 (forms 5-nk, 22-t, 2-attestation).	45	Open	Open
750	1979	Annual reports and statistical data on personnel records for 1979 (forms 5-nk, 22-t, 2-attestation).	93	Scaled	Open
820	1980	Statistical reports on personnel of the Central Economic-Mathematical Institute for 1980 (forms 5-nk, 9, 9-sr, 22-t, 1-sr (science), 1-migration).	68	Scaled	Open
920	1981	Statistical reports on personnel records of the Central Economic-Mathematical Institute for 1981 (forms 5-nk and its annexes, 8-6, 22-t).	43	Open	Open
966	1982	Statistical reports on personnel records of the Central Economic-Mathematical Institute for 1982 (forms 5-nk, 22-t) and annexes thereto.	39	Open	Open
1056	1983–1984	Statistical reports of the Central Economic-Mathematical Institute on personnel records for 1983 and 1984 (forms 5-nk, 1-k, 22-m, 9, 9-sr, 2).	79	Scaled	Scaled
1107	1985	Statistical reports on personnel records of the Central Economic-Mathematical Institute for 1985 (forms 5-nk, 1-k, 1-rs, 9, 9-sr, 2-attestation, 22-t).	45	Open	Open
1142	1986	Statistical reports on personnel records of the Central Economic-Mathematical Institute for 1986 (forms 5-nk, 1-k, kn, 6-m, 22-m).	49	Open	Open
1185	1988	Statistical reports on personnel of the Central Economic-Mathematical Institute for 1988 (forms 5-nk, 1, 9, 6-t, 22-t).	37	Open	Open

Stage 1 — Archival workbook (Russian original)

Located in demographic db(new)/. 23 files named "1959 1 X отчет.xlsx" where X is the ARAN delo number. Each file is the verbatim digital transcription of the original Russian отчет (annual report) preserved in the State Archive of the Russian Academy of Sciences (ARAN), F.1959 / Op.1 (CEMI fonds). Total 100 sheets across the 23 files.

Stage 2 — OCR intermediate (Russian)

Located in OCR Results/. 25 files named "cemi_1959_1_X_YYYY_(full|earliest|pre_cemi).xlsx". Each file is an OCR-cleaned Russian transcript of the stage-1 archival source with structured sheet layout (_Summary / _CrossReferences / Form_5nk / Form_9 / List_1 / List_2 / etc.). Filename embeds (archnum, year) for join-back to the archival original. Total 282 sheets.

Stage 3 — English translation

Located in English Translation Results/. 25 files mirroring the OCR set with suffix "_EN.xlsx". Each sheet is a one-to-one English translation of the matching OCR sheet, preserving the cell layout. Total 283 sheets.

Stage 4 — Aggregate input workbook

cemi_translation_db.xlsx consolidates the per-archival-file aggregates into a single canonical workbook with eight per-dimension sheets (Personnel Totals, Positions, Academic Degrees, Nationality, Party Membership, Age Distribution, Research Trainees) plus the Translation Glossary and Source Files Index. This is the file consumed by `load_aggregates()`.

Data Coverage and Gaps

Because the database integrates two distinct source streams — individual-level event facts taken from ARAN F.1959 / Op.1 personnel spiski and aggregate annual demographic facts taken from CEMI annual reports — its coverage is best described as a two-track inventory. Each track has its own year span, its own pattern of gaps, and its own gap aetiology (archive access restrictions versus the absence of a source form for a given year). This chapter documents both tracks and the per-table year coverage that underwrites them. Per-table coverage in detail is enumerated in Table 3 of the long version of the data-paper Methods section.

Track B — event facts (person-level)

Track B records individual career events: appointment, dual / part-time appointment, transfer, dismissal, party membership, dissertation defense, academic-title award, start of work, and the per-year person $\times$ position $\times$ department $\times$ specialty $\times$ degree observation row. Source: ARAN F.1959 / Op.1 personnel spiski (the data.xlsx workbook).

- Nominal coverage: 1961–1987. Person-year observations are populated for 25 of the 27 calendar years in this span — every year between 1961 and 1982 carries at least 22 rows and at most 1,084 rows, then 1985–1987 carry 268–504 rows each.
- ARAN access-restriction gap: 1983 and 1984. The personnel spiski packets for these two years are not yet open under ARAN's current access policy and are therefore absent from this deposit. As a result, `person_year_observation`, `cemi_appointment`, `dual_position`, `transfer_event`, `dismissal_event` and `party_event` hold 0 rows for 1983 and 1984.
- Life-event predecessor years: 1935, 1940, 1942 and 1958. Four years before the nominal 1961 start of Track B are present in the database because they are referenced by life events recorded in later spiski — a 1935 doctoral defense (`phd_defense.defense_date = "1935-01-01"`), a 1940 and a 1942 party-membership year (`party_event.join_year`), and five 1958 start-of-work dates (`start_of_work.start_date = "1958-..."`) for persons who entered the predecessor laboratory in that year. These rows do not constitute event records for those years — they are anchor dates attached to a person's CEMI-era observation row, and they are flagged by `calendar_year.institute_phase = "Pre-Institute"` (1940, 1942) or "Predecessor Lab" (1961, 1962, with 1958 inferred from `start_of_work`).
- Retrospective continuations carried in the 1985 spiski. The few Track B rows that nominally bear 1983 or 1984 dates are not newly-released event records but retrospective continuations: `defense_date = 1984-...` rows in `phd_defense` (18 rows, of which 16 are anchored to `observation_id` values whose source row sits in the 1985 packet — the defenses occurred in 1984 but are documented only via the released 1985 personnel form), and a small number of appointment / dual-position spans whose values for 1983 and 1984 are extrapolated from the adjacent 1982 and 1985 personnel rosters. Strictly event-oriented tables that do not

allow this kind of extrapolation — `transfer_event`, `dismissal_event` and the dated cells of `phd_defense` routed to non-1985 source rows — therefore show 0 rows for the 1983–1984 window.

Track A — aggregate facts (institute-level)

Track A records annual institute-wide demographic aggregates: personnel totals, position breakdowns, academic-degree composition, nationality, party membership, age distribution, research field / subfield breakdowns and research-trainee counts. Source: CEMI internal annual reports consolidated in `cemi_translation_db.xlsx` (the `demo_*` tables).

- Nominal coverage: 1961–1987.
- ARAN non-release gaps: 1979, 1980, 1983 and 1984. The corresponding annual-report packets were not released to the deposit (1979 / 1980 are non-release per the ARAN finding aid; 1983 / 1984 share the access-restriction gap with Track B). In this release every `demo_*` table — `demo_personnel_totals`, `demo_academic_degrees`, `demo_position`, `demo_field`, `demo_subfield`, `demo_nationality`, `demo_party`, `demo_age` — carries 0 rows for 1979, 1980, 1983 and 1984.
- Selective single-year tables. Two demographic tables have substantially sparser coverage than the others by the design of the source form, not by archive restriction: `demo_age` carries snapshot rows only for 1966, 1972, 1977, 1982 and 1987 (the annual reports that include age-distribution breakdowns) and `demo_nationality` is populated only for the subset of years for which the report transcribes a nationality breakdown. These are source-form gaps and must be interpreted differently from ARAN access-restriction gaps (see below).

Two distinct gap aetiologies

The codebook draws an explicit distinction between two kinds of zero-row cells in the per-table year-coverage table. Confusing the two leads to systematic analytical error.

- ARAN access-restriction gap. The archival packet exists and is catalogued in the F.1959 / Op.1 finding aid, but is not currently open to researchers under ARAN's access policy. No row exists in the database for that (table, year) cell. This is the situation in 1983 and 1984 for both Tracks A and B and in 1979 and 1980 for Track A. When such years are reopened in a future release the corresponding rows will simply be appended; no schema migration is required.
- Source-form gap. The packet for that year is open and present in the deposit, but the source form does not record the field at all (e.g. the 1965 annual report does not include an age-distribution breakdown). No row exists in the database for that (table, year) cell either, but for a fundamentally different reason: there is nothing to release. This is the situation that affects `demo_age` (rows for 1982 and 1987 only) and partially `demo_nationality`.

For analytical purposes ARAN access-restriction gaps should be treated as missing-at-random with respect to the analysis (subject to the user's own judgment) and may license imputation against the adjacent open years. Source-form gaps must be treated as structurally missing — no imputation against adjacent years is appropriate because the field was never collected for the gapped year.

Per-table year coverage summary

The following per-table summary lists the year span actually populated in each Track B event table and each Track A aggregate table, together with the explicit gap years. The numbers refer to row counts in the live `cemi_career.db` (full-fidelity build) at the date of this release; the year spans and gap years are identical in both build tiers.

Table	Track	Populated year span	Gap years
person_year_observation	B	1961–1987	1983, 1984 (ARAN access-restriction)
cemi_appointment	B	1961–1987	1983, 1984 (ARAN access-restriction)
dual_position	B	1961–1987	1983, 1984 (ARAN access-restriction)
transfer_event	B	1965–1987	1983, 1984 (ARAN access-restriction)
dismissal_event	B	1963–1987	1983, 1984 (ARAN access-restriction)
phd_defense	B	1935; 1955–1987 (sparse before 1965)	1983 only (1984 carries 18 retrospective rows via the 1985 spiski)
party_event (join_year)	B	1940; 1942; 1961–1987	1983, 1984 (ARAN access-restriction)
start_of_work	B	1958–1987	1983, 1984 (ARAN access-restriction)
academic_title_award	B	1961–1987	1983, 1984 (ARAN access-restriction)
pre_cemi_role	B	— (no year column; linked via observation_id)	— (inherits gaps of person_year_observation)
demo_personnel_totals	A	1961–1987	1979, 1980, 1983, 1984 (ARAN non-release)
demo_academic_degrees	A	1961–1987	1979, 1980, 1983, 1984 (ARAN non-release)
demo_position	A	1961–1987	1979, 1980, 1983, 1984 (ARAN non-release)
demo_field	A	1961–1987	1979, 1980, 1983, 1984 (ARAN non-release); 1986 source-form gap
demo_subfield	A	1961–1987	1979, 1980, 1983, 1984 (ARAN non-release); 1986 source-form gap

Table	Track	Populated year span	Gap years
demo_nationality	A	selective years	multiple source-form gaps + 1979, 1980, 1983, 1984 ARAN gaps
demo_party	A	1961–1987	1979, 1980, 1983, 1984 (ARAN non-release)
demo_age	A	1966, 1972, 1977, 1982, 1987	source-form gap (other years did not transcribe age brackets)
demo_trainees	A	selective years	source-form gap (training counts only transcribed selectively)

Analytical implications

Two practical implications for analysts working with the database.

- Yearly trend computations over Track B series should be normalised to the open-year set {1961, ..., 1982, 1985, 1986, 1987} = 25 years. Including 1983 and 1984 in a denominator as if they were zero-event years will systematically understate the rate of any event type. When a denominator must span the full 1961–1987 nominal coverage, analysts should explicitly flag 1983 and 1984 as restricted-access and exclude them from the time-series operation.
- Joining Tracks A and B by year should be done on the intersection of their open-year sets — that is, on {1961, ..., 1978, 1981, 1982, 1985, 1986, 1987} = 23 years — because Track A has additional non-release gaps in 1979 and 1980 that Track B does not share. Any join over the full 1961–1987 nominal span will create empty join rows for 1979, 1980, 1983 and 1984 and must be filtered explicitly.
- The few 1984 phd_defense rows must be interpreted as retrospective documentation, not as evidence that 1984 is open: the defense happened in 1984 but is recorded by the 1985 spiski, the same source that recorded the analyst's own 1985 employment row. Counting these rows as "1984 evidence" overstates 1984 coverage and undercounts 1985 source utilisation. A SQL audit join from phd_defense.observation_id back to person_year_observation.year recovers the correct source year of each row.

References

A. Historical-database methodology

These works supply the methodological frame that this codebook implements — conceptual modeling of historical records, intermediate / event-oriented data structures, persistent-identifier construction, factoid prosopography, source-criticism and provenance preservation, and longitudinal-population-database practice.

- Harvey, C. & Press, J. *Databases in Historical Research: Theory, Methods and Applications* (Macmillan Education UK, London, 1996). — foundational text; Ch. 4. The Database Project Life-Cycle(source criticism / data capture) and Ch. 5. Database Design and Implementation (codebook design and variable-block format) directly inform the structure of this document.
- Gil, T. L. *How to Make a Database in Historical Studies* (Springer, 2021). — practical handbook covering archive-to-schema design (Ch. 1. Some Theoretical and Methodical Questions), data-entry principles (Ch. 2. Some Computer Issues), variable-block codebook conventions (Ch. 3. Research Engineering), and reproducibility (Ch. 4. Constructing Databases: Some Concrete Examples).
- Mandemakers, K., Alter, G., Vézina, H. & Puschmann, P. (eds) *Sowing: The Construction of Historical Longitudinal Population Databases* (Radboud University Press, 2023). — methodological standard for historical longitudinal population databases; the design template for event-based observation records.
- Alter, G. & Mandemakers, K. The Intermediate Data Structure (IDS) for Longitudinal Historical Microdata, version 4. *Historical Life Course Studies* **1**, 1–26 (2014). — formalises the event-based "(person, date, event-type, value)" data model that underpins this database's appointment / transfer / dismissal / dual-position tables.
- Pasin, M. & Bradley, J. Factoid-based Prosopography and Computer Ontologies: Towards an Integrated Approach. *Digital Scholarship in the Humanities* **30**, 86–97 (2015). — factoid model adapted here for person-year personnel observations (every career fact recorded as a discrete factoid with explicit source attribution).
- Dedieu, J.-P. *Designing Databases for Historical Research – With Special Reference to Fichoz*. Working paper (ResHist – Réseaux & histoire, OpenEdition Hypothèses, 2016); https://reshist.hypotheses.org/files/2016/07/Dedieu_Historical_Databases.pdf. — Fichoz schema (person-event-source triplet) influencing the linkage between person, calendar_year, and event tables.
- Akoka, J., Comyn-Wattiau, I., Lamassé, S., & du Mouza, C. (2020). Contribution of Conceptual Modeling to Enhancing Historians' Intuition – Application to Prosopography. In G. Dobbie, U. Frank, G. Kappel, S. W. Liddle, & H. C. Mayr (Eds.), *Conceptual Modeling. ER 2020* (Lecture Notes in Computer Science, Vol. 12400, pp. 164–173). Springer, Cham. https://doi.org/10.1007/978-3-030-62522-1_12 — conceptual-

modeling approach for prosopographical databases; cited here as the rationale for the three-level (Person / Person×Year / Event) hierarchy used in §Data Units.

- Gutmann, M. P., Fliess, K. H., Holmes, A. E., Fairchild, A. L., & Teas, W. A. (1989). Keeping Track of our Treasures: Managing Historical Data with Relational Database Software. *Historical Methods: A Journal of Quantitative and Interdisciplinary History*, 22(4), 128–143.
<https://doi.org/10.1080/01615440.1989.10594186> — early statement of provenance-preserving relational-DB construction for historical research, cited here for the source_material backlinking convention.
- Murata, T. (2022). The Construction of the Historical Disaster Evidence Database and its Effectiveness. *Journal of Disaster Research*, 17(3), 420–429 (2014) <https://doi.org/10.20965/jdr.2022.p0420>. — comparable single-institute heritage-evidence DB cited here for the per-record evidence/provenance-string convention used in institution.classification_evidence.
- Boonstra, O., Breure, L. & Doorn, P. *Past, Present and Future of Historical Information Science* 2nd edn (DANS – Data Archiving and Networked Services, KNAW, Amsterdam, 2006). — survey of historical-information-science methodology; situates the present database within that scholarly tradition.

B. Conceptual modeling, data quality, and consistency

These works supply the formal/theoretical apparatus for the conceptual schema (CIDOC-CRM), general relational-database design conventions, and the inconsistency-tolerant data-quality approach used in the technical validation of the database.

- Bekiari, C., Bruseker, G., Canning, E., Doerr, M., Michon, P., Ore, C.-E., Stead, S., & Velios, A. (Eds.). *Definition of the CIDOC Conceptual Reference Model* (Version 7.1.3, Volume A). CIDOC CRM Special Interest Group. (2024) <https://www.cidoc-crm.org/> (Appointment, Transfer, Dismissal, etc.).
- Foster, E. C. & Godbole, S. V. *Database Systems: A Pragmatic Approach* 3rd edn (CRC Press, 2023). — general relational-database-design textbook reference for the relational schema (3NF normalisation in core tables, controlled denormalisation in derived tables such as cemi_position_span and person_degree); the null-value semantics underpinning the “missing in source” vs “explicitly unknown” distinction follow Codd’s Rule 3 (Systematic Treatment of Null Values), Ch. 9 §9.2.2.2.
- Afrati, F. N. & Kolaitis, P. G. Repair Checking in Inconsistent Databases: Algorithms and Complexity. In *Proceedings of the 12th International Conference on Database Theory (ICDT 2009)* 31–41 (ACM, 2009). <https://doi.org/10.1145/1514894.1514899> — theoretical foundation for treating manual classification_evidence overrides as repairs over a primary-key-constrained relation.
- Wijzen, J. Consistent Query Answering under Primary Keys: A Characterization of Tractable Queries. In *Proceedings of the 12th International Conference on Database Theory (ICDT 2009)* 42–52 (ACM, 2009). <https://doi.org/10.1145/1514894.1514900> — formal basis for the audit query "every kind assignment must carry a non-NULL classification_evidence" used in technical validation.

- Vianu, V. Automatic Verification of Database-driven Systems: A New Frontier. In *Proceedings of the 12th International Conference on Database Theory (ICDT 2009)* 1-12 (ACM, 2009). — frames the build pipeline's assertion suite as an automatic-verification problem over a relational state machine.

C. Soviet R&D-institution classification (primary and secondary sources)

Detailed per-sector citations appear under §"Institution Type Classification" → "References specific to this classification". Summary listing including the additional secondary-source corpus consulted for boundary cases:

- Nolting, L. E. & Feshbach, M. R&D Employment in the U.S.S.R. *Science* 207(4430), 493–503 (1980). — primary 4-sector taxonomy article.
- Nolting, L. E. & Feshbach, M. *Statistics on Research and Development Employment in the U.S.S.R.* Series P-95, No. 76 (U.S. Bureau of the Census, 1981). — primary and book-length elaboration of the 4-sector taxonomy with per-sector universe definitions, used as the canonical operational reference for Academy / Branch / VUZ / Enterprise.
- Nolting, L. E. *Financing of Research, Development, and Innovation in the U.S.S.R.* Foreign Economic Report No. 22 (U.S. Bureau of the Census, Washington, D.C., 1985). — financing-side counterpart to the 1981 P-95 No. 76 monograph; companion to the sector-coverage definitions.
- Graham, L. R. The formation of Soviet research institutes: a combination of revolutionary innovation and international borrowing. *Social Studies of Science* 5, 303–329 (1975). — historical basis for the VUZ-sector treatment of higher-education research and the AN-vs-VUZ academy distinction.
- Cocks, P. M. *Science Policy: USA/USSR*. Vol. II: Science Policy in the Soviet Union. Report NSF/PRA-NS-80-09 (National Science Foundation, 1980). — secondary support for the Branch / Enterprise / Academy sector boundaries, including Ch. VIII on ministry-line ("Branch") R&D organisation.
- Kassel, S. & Campbell, C. *The Soviet Academy of Sciences and Technological Development*. Report R-2533 (RAND Corporation, Santa Monica, CA, 1980). — primary reference for the Academy / Branch interface in defence-related R&D, used here to classify ambiguous Academy-Branch boundary cases.
- Central Intelligence Agency. *Organization of Soviet Science*. Report CIA/RR ER-59-22 (CIA-RDP79R00904A000500010075-6) (1959). — earliest English-language sector-by-sector inventory of Soviet R&D institutions; supplies historical baseline for the 4-sector taxonomy.
- Gerovitch, S. *Voices of the Soviet Space Program: Cosmonauts, Soldiers, and Engineers Who Took the USSR into Space* (Palgrave Macmillan, 2014). — supplementary glossary for branch / enterprise classification of space-sector institutes (КБ / ОКБ / ЦКБМ etc.).
- Dexter, K. & Rodionov, I. *The Soviet Defence Industry Enterprise Database* (Warwick Centre for the Study of Globalisation and Regionalisation); consolidated workbook + per-entity preface PDFs (versions 5–22) (2007–2017). — primary reference for enterprise / branch classification of defence-industry institutions

(НИИ, КБ, ОКБ, ЦКБ, СКБ, ГСПИ); the per-entity preface PDFs were consulted for individual classification decisions.

- USSR Central Statistical Administration (ЦСУ СССР). *Narodnoye khozyaystvo SSSR za 60 let (Народное хозяйство СССР за 60 лет) / 1977: metodologiya (методология)* (Statistika, 1977). — TsSU statistical methodology yearbook; defines the official Soviet S&T statistical universe that the Nolting 4-sector taxonomy operationalises.
- USSR Central Statistical Administration (ЦСУ СССР). *Narodnoye obrazovaniye, nauka i kul'tura v SSSR: statisticheskiy sbornik (Народное образование, наука и культура в СССР: статистический сборник)* (Statistika, 1977). — companion statistical compendium covering the VUZ-sector and Research Institutions.
- USSR Council of Ministers. Decree No. 760 of 24 September 1968 «O sozdanii proizvodstvennykh i nauchno-proizvodstvennykh ob"yedineniy (О создании производственных и научно-производственных объединений)». — formalised the NPO (Scientific-Production Association) status that pins the Branch / Enterprise boundary.
- USSR Council of Ministers. Decree of 13 June 1961 «O merakh po dal'neyshemu razvitiyu nauchno-pedagogicheskikh kadrov v vysshikh uchebnykh zavedeniyakh SSSR (О мерах по дальнейшему развитию научно-педагогических кадров в высших учебных заведениях СССР)». — formalised the VUZ-sector R&D funding stream.